

Strict Cone Separation from Positive KKT Certificates: a Gordan–Farkas Mechanism and a Local Sharpness Theorem

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Abstract

We isolate a reusable mechanism that converts a *strictly positive* KKT certificate into a strictly quantitative separation statement, and we determine its sharp constant in a concrete non-smooth extremal problem. Let $G \in \mathbb{R}^{m \times n}$ admit $y > 0$ with $G^\top y = 0$. Gordan’s theorem of alternatives then forces the descent cone to collapse, $\{h : Gh \leq 0\} = \{0\}$, hence $\text{rank } G = n$, and the column space contains no nonzero nonnegative vector; consequently the augmented matrix $[G \mid -e]$ with $e \geq 0$, $e \neq 0$ is nonsingular. For the associated linear program $\min c$ subject to $v_i h \leq -1$ and $w_q h \leq c$ we prove $c_* = 1/L$ with $L = \sum_q \lambda_q$, where λ are the positive multipliers of $G^\top y = 0$; the dual-optimal Farkas certificate is exactly $c_*(\omega, \lambda)$. Quantitatively, the sharpness constant $\eta = \min_{\|u\|=1} \max_i (Gu)_i$ is certified strictly positive ($\eta \geq 0.0766443030678\dots$ in the instance), and a second-order Taylor bound with explicit constant $K = 648$ yields a certified local radius $\rho_{\text{cert}} = \eta/K = 1.18278 \times 10^{-4}$ inside which the local maximum strictly exceeds its boundary value. Finally we characterise the *scope boundary* of the mechanism: the certificate is a positive combination and therefore belongs to the sum-of-nonnegative type; canonical involution-invariant data cannot yield the one-sided, axis-breaking separator that an arithmetic realisation would require, and the orientation available at the membership boundary we examine is Archimedean in origin. Thus the mechanism is a strictly proved *local geometric* separation statement, and it does not by itself upgrade to an arithmetic or spectral one. No claim about the Riemann Hypothesis is made or used. The resulting negative statement is deliberately formulated as a *registered-framework obstruction*: within the canonical framework we specify, non-vacuity and one-sided discrimination are incompatible, and we do not claim a complete impossibility theorem; instead we isolate precisely the single unexcluded degree of freedom (a non-canonical, ι -breaking Archimedean object).

1 Setting and hypotheses

Fix $n = 5$, $m = n + 1 = 6$, and a sign vector $\sigma = (\sigma_1, \dots, \sigma_5) \in \{\pm 1\}^5$. For $\phi = (\phi_1, \dots, \phi_5) \in (0, \pi/2)^5$ put

$$F_k(\phi) = \sum_{j=1}^5 \gamma_j \cos(k\phi_j), \quad \gamma_j = \begin{cases} \sigma_j, & k \text{ odd}, \\ 1, & k \text{ even}, \end{cases} \quad (1)$$

so that F_k is a trigonometric polynomial of degree k . Let ϕ^* be a point at which the *six-active* conditions hold for the odd frequencies 13, 19 and the even functionals with $2q \in \{6, 8, 14, 18\}$, i.e. frequencies 12, 16, 28, 36. Define the signed active functions

$$g_{13} = -F_{13}, \quad g_{19} = F_{19}, \quad g_q = F_q - \frac{1}{2} \quad (q \in \mathcal{A} := \{6, 8, 14, 18\}). \quad (2)$$

Write $G \in \mathbb{R}^{6 \times 5}$ for the matrix whose rows are the gradients $\nabla g_i(\phi^*)$, ordered as (13, 19, 6, 8, 14, 18).

- (H1) (*strict positive KKT certificate.*) There is $y \in \mathbb{R}^6$ with $y > 0$ componentwise and $G^\top y = 0$; we write $y = (\omega_{13}, \omega_{19}, \lambda_6, \lambda_8, \lambda_{14}, \lambda_{18})$ and $L = \sum_{q \in \mathcal{A}} \lambda_q > 0$.
- (H2) (*normalisation.*) $\omega_{13} + \omega_{19} = 1$.
- (H3) (*attainment of the active square system.*) There exist $h^* \in \mathbb{R}^5$ and $c^* \in \mathbb{R}$ with $v_i h^* = -1$ for $i \in \{13, 19\}$ and $w_q h^* = c^*$ for all $q \in \mathcal{A}$, where v_i, w_q are the corresponding rows of G .

In the concrete instance we use, $\sigma = (-1, 1, 1, -1, 1)$ and the numerical realisation of ϕ^* gives (to the stated precision)

$$L = 1.99676941227516035204325246457, \quad c^* = 0.500808953629041885590052265792.$$

Hypothesis (H1) is a strict-complementarity/positive-KKT hypothesis: it holds at the point considered and is *not* automatic at degenerate points (see §5).

2 The abstract mechanism

We use the following classical alternative; see e.g. Gordan's original note and standard texts.

Theorem 2.1 (Gordan, [1, 2]). *For $G \in \mathbb{R}^{m \times n}$, exactly one of the following holds: (i) there is $h \neq 0$ with $Gh \leq 0$; (ii) there is $y > 0$ with $G^\top y = 0$.*

Lemma 2.2 (Trivial descent cone). *Assume (H1). Then $\{h \in \mathbb{R}^n : Gh \leq 0\} = \{0\}$.*

Proof. By (H1), alternative (ii) of Theorem 2.1 holds; hence (i) fails, which is the assertion. $\square$

Corollary 2.3 (Structural full column rank). *Assume (H1). Then $\text{rank } G = n$.*

Proof. If $\text{rank } G < n$ then $\ker G \neq \{0\}$; choosing $0 \neq h \in \ker G$ gives $Gh = 0 \leq 0$, contradicting Lemma 2.2. $\square$

Corollary 2.4 (No nonzero nonnegative vector in the column space). *Assume (H1). Then $\text{col}(G) \cap \{z \geq 0, z \neq 0\} = \emptyset$. In particular, for every $e \geq 0, e \neq 0$, one has $e \notin \text{col}(G)$.*

Proof. Suppose $0 \neq z \geq 0$ and $z \in \text{col}(G)$. Since $\text{col}(G)$ is a linear subspace, $-z \in \text{col}(G)$ as well, so there is h with $Gh = -z \leq 0$. Moreover $h \neq 0$, because $z \neq 0$. This contradicts Lemma 2.2. $\square$

Note that Corollary 2.4 yields nonsingularity of the augmented matrix *structurally*, i.e. from the positive certificate and not from a numerical determinant:

$$m = n + 1, \quad \text{rank } G = n, \quad e \notin \text{col}(G) \implies \text{rank}[G \mid -e] = n + 1. \quad (3)$$

3 The exact bridge constant

Consider the linear program

$$c_* = \min_{h, c} c \quad \text{subject to} \quad v_i h \leq -1 \quad (i \in \{13, 19\}), \quad w_q h - c \leq 0 \quad (q \in \mathcal{A}). \quad (4)$$

Theorem 3.1 (Exact bridge constant). *Assume (H1), (H2) and (H3). Then*

$$c_* = \frac{1}{L}, \quad L = \sum_{q \in \mathcal{A}} \lambda_q.$$

Moreover the Farkas dual certificate $(\nu, \mu) = (\omega/L, \lambda/L)$ is dual-optimal and satisfies $(\nu, \mu) = c_*(\omega, \lambda)$.

Proof. Lower bound. Let h be primal-feasible. Using $v_{13}h \leq -1$, $v_{19}h \leq -1$ and (H2) with $\omega \geq 0$,

$$\omega_{13}(v_{13}h) + \omega_{19}(v_{19}h) \leq -(\omega_{13} + \omega_{19}) = -1.$$

By (H1), $\omega_{13}v_{13} + \omega_{19}v_{19} = -\sum_q \lambda_q w_q$, hence $\sum_q \lambda_q w_q h \geq 1$. Since $w_q h \leq \max_r w_r h$ and $\lambda \geq 0$, $L \max_q (w_q h) \geq \sum_q \lambda_q w_q h \geq 1$, so $c_* \geq 1/L$.

Upper bound. Let (h^*, c^*) be as in (H3) and take the inner product of $G^\top y = 0$ with h^* . The odd components contribute $-\omega_{13} - \omega_{19} = -1$ and the even components $\sum_q \lambda_q w_q h^* = c^* L$, so $-1 + c^* L = 0$, i.e. $c^* = 1/L$. By (H3), h^* is primal-feasible, whence $c_* \leq c^* = 1/L$.

Combining the two bounds gives $c_* = 1/L$. Finally, $(\nu, \mu) = (\omega/L, \lambda/L)$ satisfies $\nu, \mu \geq 0$, $\sum_q \mu_q = 1$ and $\nu_{13}v_{13} + \nu_{19}v_{19} + \sum_q \mu_q w_q = 0$, so it is dual-feasible with value $\nu_{13} + \nu_{19} = 1/L = c_*$; dual optimality follows, and the displayed identity $(\nu, \mu) = c_*(\omega, \lambda)$ is immediate. $\square$

Remark 3.2. Theorem 3.1 shows that c_* is not an isolated numerical constant: it is the reciprocal of the sum of the active even multipliers, and the Farkas separator is the KKT certificate rescaled by c_* .

4 Quantitative sharpness and a certified local radius

Define the sharpness constant

$$\eta = \min_{\|u\|_2=1} \max_i (Gu)_i. \quad (5)$$

By Lemma 2.2, $\max_i (Gu)_i > 0$ for every $u \neq 0$, and the unit sphere is compact, so $\eta > 0$; this is a second, quantitative consequence of the positive certificate.

To certify a numerical lower bound we exploit the sign decomposition: every $u \neq 0$ is uniquely $u = s \odot w$ with $s \in \{\pm 1\}^5$, $w \geq 0$, $\sum_j w_j = 1$, and $\max_i (Gu)_i / \|u\|_2 \geq \max_i (Gu)_i / \|u\|_1$. Hence $\eta_1 = \min_s \min\{\max_i (G(s \odot w))_i : w \geq 0, \mathbf{1}^\top w = 1\}$ satisfies $\eta \geq \eta_1$, and each inner problem is a linear program. By weak duality, for any y in the simplex, $\max_i (G(s \odot w))_i \geq \min_j ((G_s)^\top y)_j$, which requires only $y \geq 0$ and $\sum_i y_i = 1$. Enclosing the coefficients rationally gives the certified bound

$$\eta \geq \underline{\eta} = 0.076644303067798235521 \dots > 0. \quad (6)$$

For the Taylor step, work in the coordinates ϕ . Since $\partial^2 F_k / \partial \phi_j^2 = -k \gamma_j \cos(k \phi_j)$, the Hessian is diagonal with $|\partial^2 F_k / \partial \phi_j^2| \leq k^2$, hence for the even rows $k \leq 36$ and the odd rows $k \leq 23$, the remainder satisfies $|R_i(h)| \leq K \|h\|_2^2$ with

$$K = \frac{1}{2} \cdot 36^2 = 648. \quad (7)$$

Writing $g_i(\phi^* + h) = g_i(\phi^*) + \nabla g_i(\phi^*) \cdot h + R_i(h)$ and using $g_{13}(\phi^*) = g_{19}(\phi^*) = c_0$, $g_q(\phi^*) = 0$, we obtain for every $h \neq 0$

$$\max_i g_i(\phi^* + h) \geq c_0 + \eta \|h\|_2 - K \|h\|_2^2. \quad (8)$$

Consequently

$$\boxed{0 < \|h\|_2 < \rho_{\text{cert}} := \frac{\eta}{K} \implies \max_i g_i(\phi^* + h) > c_0} \quad (9)$$

and in the instance at hand $\rho_{\text{cert}} = 1.18278 \times 10^{-4}$ (the tightest coordinate half-width in the original variable is 7.99×10^{-5}).

Remark 4.1. Statement (9) is a *local* sharpness theorem: it certifies that ϕ^* is a strict local minimiser of the local maximum, inside a fully explicit neighbourhood. It does not assert global optimality; in particular no equality $V = c_0$ is claimed for the global problem.

5 Scope of the mechanism: an interface obstruction

The mechanism above is a statement of linear-algebraic/convex duality. We record honestly why it does not automatically upgrade to an arithmetic or spectral statement.

Proposition 5.1 (Class of the certificate). *The certificate in (H1) is a positive combination, $\sum_i y_i g_i = 0$ with $y > 0$. Consequently the mechanism belongs to the sum-of-nonnegatives positivity type, and renaming it (e.g. as a “Farkas separation” or “positive KKT certificate”) does not make it a new positivity mechanism.*

Proof. Immediate from the displayed identity, which is exactly (H1) rewritten row-wise. $\square$

Proposition 5.2 (Symmetry obstruction). *Suppose the available arithmetic data are invariant under the involution ι associated with the functional equation (so that condition sets are ι -invariant). Then no one-sided, axis-breaking separator can be produced from them: a symmetric datum cannot generate an asymmetric (one-sided) separation object.*

Proof. If a predicate P satisfies $P(\beta) = P(\iota\beta)$ for all admissible data, then it cannot separate β from $\iota\beta$; a one-sided separator by definition takes different values on the two sides, so it cannot be realised by such P . $\square$

Proposition 5.3 (Archimedean orientation at the membership boundary). *For the membership boundary we examined, the available orientation is Archimedean in origin: the comparison involves an Archimedean constant and an Archimedean iterated logarithm, so the resulting direction is not arithmetic.*

Proof. Direct inspection of the comparison defining the boundary: the two sides involve $\sigma(n)$ and $e^\gamma n \log \log n$ respectively, and e^γ , $\log \log n$ are Archimedean objects. $\square$

Proposition 5.4 (Incompatibility of non-vacuity and one-sided discrimination). *Suppose that (i) every non-vacuous balancing factor for the involution ι is Archimedean, (ii) the Archimedean factors available in the canonical framework are universal (they depend only on the completion type, not on the zero configuration of the object, and are therefore β -blind), and (iii) producing a separator whose output is one-sided in the ι -variable requires data that are not ι -invariant. Then no separator can be simultaneously non-vacuous and one-sided: the two requirements are mutually exclusive within this framework.*

Proof. By (i) a non-vacuous balancing factor is Archimedean; by (ii) it is universal, hence β -blind and unable to discriminate β -values; by (iii) discrimination requires data that vary with the object. Contradiction. Contradiction. $\square$

Remark 5.5. Proposition 5.4 isolates the remaining degree of freedom to a non-canonical, ι -breaking Archimedean object. We neither construct such an object nor assert its existence or non-existence; the statement is about the framework, not about the objects alone.

Remark 5.6 (Scope boundary). Together, Propositions 5.1–5.4 delimit the scope of the mechanism: it is a strictly proved *local geometric* separation statement with an exact constant, and the passage to arithmetic/spectral content requires an interface that is not supplied here. We state the resulting separation as a scope boundary,

$$\text{local arithmetic structure} \not\Rightarrow \text{global spectral localisation},$$

rather than as a claim about any particular spectral problem. No Riemann Hypothesis statement is used or asserted anywhere in this paper.

6 Data and reproducibility

All numerical statements are accompanied by scripts and machine-checked outputs in the accompanying repository: the reconstruction of ϕ^* at high precision, the six-active gradient matrix G , the Gordan certificate y , the 32-sign linear programs certifying (6), and the Taylor/remainder computation. The certified lower bound $\underline{\eta}$ is obtained from dual feasible points (only $y \geq 0$, $\sum_i y_i = 1$ are used) together with a rational enclosure of the coefficients; the margin used in the enclosure is 10^{-30} , against a truncation error of order 10^{-45} at the working precision.

References

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